AI Work Verification and Auto-Invoicing Platform

Strategic Product Document

Build a platform that cross-verifies work completed against hours logged, compares delivery effort against internal and later market benchmarks, validates invoices, and auto-generates invoice drafts from approved work records.

Prepared for team alignment, planning, and sharing with AI/copilot systems.

1. Executive Summary

This product will help service teams answer three important questions:

1. Was the logged work supported by real delivery evidence?
2. Was the time and cost reasonable compared to similar internal work and future market benchmarks?
3. Can that verified work flow into faster, more accurate invoice creation?

The fastest route to value is a tight first release focused on PM tools, time tracking, verification logic, and invoice draft generation. Wider AI analysis and deeper benchmarking can then expand during beta and after customer feedback.

Primary Problem

Hours are logged in one tool, delivery happens in another, and invoices are created elsewhere. This creates weak visibility, billing disputes, and poor benchmark intelligence.

Initial Solution

Connect PM, time, and accounting systems; verify hours against work signals; flag anomalies; and generate invoice drafts from approved records.

2. Product Vision

Working statement: An AI-assisted work verification and invoicing platform that turns task activity, time logs, and business records into verified, benchmarked, invoice-ready output.

- Reduce unsupported or inflated time entries
- Improve billing confidence
- Speed up invoice preparation

- Reduce disputes and manual reconciliation
- Create a long-term benchmark layer across teams and projects

3. First Target Users

- Agencies
- Software consultancies
- Professional services teams
- Delivery, operations, and finance leaders inside service-based businesses

4. Core Product Capabilities

Capability	What it does
Work Verification	Links time entries to tasks and supporting activity, then scores whether work appears sufficiently evidenced.
Benchmarking	Compares effort and cost to similar internal work first, then later to external market norms.
Invoice Validation	Checks invoice lines, rates, users, and hours against approved work and system records.
Auto-Invoicing	Creates invoice drafts from approved users, tasks, hours, and rate card rules.

5. Phase-by-Phase Delivery Plan

PHASE 1 - FASTEST RELEASE / MVP

Main Goal

Ship a focused product quickly and get into beta as fast as possible.

Phase 1 Product Promise

Connect your PM and time systems, verify whether logged work is supported by delivery activity, and generate invoice drafts from approved records.

Scope

- PM integration: Jira or ClickUp
- Time-tracking integration: Harvest or Tempo

- Accounting integration: QuickBooks or Xero
- Optional support signal: GitHub
- Verification dashboard
- Invoice draft generator
- Review and approval flow

Must-have Features

1. Organization/workspace setup
2. User identity mapping across systems
3. Project, task, user, and time entry sync
4. Rules-based work verification
5. Verification dashboard for flagged and approved records
6. Invoice draft generation from approved hours
7. Shareable invoice support summary

AI in Phase 1

- Summarize verified work for invoice descriptions
- Explain why a time entry was flagged
- Help normalize unclear task descriptions

Out of Scope

- Deep email parsing
- Invoice OCR
- External market benchmarking
- Complex ERP workflows
- Fully automated invoice sending

Why this phase first: It is the fastest path to real customer value, clean beta feedback, and a strong product story without overbuilding too early.

Phase 1 Exit Criteria

- A customer can connect tools successfully
- A manager can review verified versus flagged hours
- A finance user can generate a draft invoice from approved work
- The system explains why entries were flagged

PHASE 1 ADD-ON - PITCH DECK STYLE VERSION

Slide 1 - Title: AI Work Verification and Auto-Invoicing Platform. Verify work. Validate hours. Generate invoices faster.

Slide 2 - Problem: Work happens across multiple tools, creating weak proof of delivery and invoice friction.

Slide 3 - Solution: Connect PM tools, timesheets, and accounting systems to verify, flag, and invoice from approved work.

Slide 4 - Why now: AI can explain work evidence, while finance teams need auditability and faster billing workflows.

Slide 5 - Target customer: Agencies, consultancies, and professional services organizations.

Slide 6 - MVP: Jira/ClickUp + Harvest/Tempo + QuickBooks/Xero + verification dashboard + invoice draft generation.

Slide 7 - Value: Faster invoicing, fewer disputes, better billing confidence, and improved accountability.

Slide 8 - Moat: Cross-system mapping, verification logic, historical benchmarks, and finance-delivery alignment.

Slide 9 - Business model: SaaS pricing by workspace/team size, with premium analytics and controls.

Slide 10 - Near-term roadmap: Ship fast, start beta, then expand into OCR, email, and benchmarking.

PHASE 2 - BETA EXPANSION

Main Goal

Increase trust, validation depth, and real-world coverage while early customers actively use the MVP.

Scope

- Gmail and Outlook integration
- Invoice PDF upload
- OCR/document extraction
- Cross-check invoice lines against approved work
- Basic internal benchmarking by task type, role, and project
- Exception review queue

Beta Focus

- Accuracy of verification scores
- Usefulness of discrepancy flags
- Cross-system mapping quality
- Finance team confidence in draft invoices

Exit Criteria

- Beta teams trust the system enough to review it regularly
- Invoice validation catches real mismatches
- Internal benchmark views are useful for decision-making

PHASE 3 - INTELLIGENCE AND BENCHMARKING

Main Goal

Move from operational verification into delivery intelligence and performance analysis.

- Historical benchmarking engine
- Cross-team and cross-project comparisons
- Role and project-level productivity trends
- Effort variance analysis
- Margin and efficiency insights

PHASE 4 - MARKET BENCHMARKING AND ADVANCED AUTOMATION

Main Goal

Add external market context and deeper automation once internal data quality and auditability are strong.

- External market rate benchmarks
- Pricing recommendations
- Contract-aware billing suggestions
- Auto-approval suggestions with human review
- Advanced finance workflow automation

6. Recommended Tech Stack

Layer	Recommendation
Frontend	Next.js, TypeScript, Tailwind CSS, shadcn/ui
Core backend	NestJS with TypeScript
AI/analytics service	Python + FastAPI, pandas, scikit-learn, LLM APIs
Database	PostgreSQL, optionally pgvector
Jobs and workflows	Redis + BullMQ or Temporal
Files and documents	AWS S3
OCR/document extraction	AWS Textract or Google Document AI
Auth	Auth0, Clerk, or WorkOS
Infra/observability	AWS, containers, OpenTelemetry, Datadog or Grafana

7. Architecture Principles

- **Deterministic rules first:** rely on source data and rules before advanced AI judgement.
- **AI as assistant:** use AI for explanation, summarization, and support rather than as sole decision-maker.
- **Full audit trail:** every result must show what data, rule, model, and approver were involved.

- **Normalized entities:** standardize users, clients, projects, tasks, time entries, invoices, invoice lines, rate cards, and evidence events.

---

8. Recommended Beta Metrics

- Percentage of time entries mapped to tasks
- Percentage of hours verified versus flagged
- Invoice draft generation time
- Number of mismatches caught before invoicing
- Human overrides of flagged entries
- Finance review time per invoice cycle

---

9. Key Open Questions

1. Who is the exact first customer profile?
2. Is the biggest pain point verification, invoicing, or benchmarking?
3. Should Phase 1 create invoice drafts only, or also send them?
4. Which PM, time, and accounting integrations matter most at launch?
5. How should pricing work: by seat, workspace, invoices, or usage?
6. How deep should email and document permissions go?
7. Where will market benchmark data eventually come from?

---

10. Final Recommendation

Move fast with a tightly scoped first release centered on PM task sync, time-log verification, approval workflows, and invoice draft generation. Use beta usage to harden the verification engine and guide the rollout of OCR, email validation, and benchmarking in subsequent phases.